Technical Notes and Reproduction Details

A Path Is Not a State: Exact Predictive Refinement of Fixed Connectome Partitions

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These notes retain finite-domain definitions, historical counts, and replay instructions supporting the research report. They do not enlarge the theorem domain or replace the published evidence packages of Papers XVI and XVII.

T1. Finite Domains

Table T1. Distinct operations and their checking data.

Operation	Finite domain	Obstruction and checking data
Nonnegative depth-two route	Full-release edge table; three annotation policies	Disjoint incoming and outgoing relay sets; complete relay scan
Signed routed words	Signed normalized carrier; somaSide and superclass	Exact routed-product zero on a supported word
Unit-source one-step prediction	163,439 eligible sources; sector means	Unequal positive outgoing sector profiles in one observation fiber
Reached-set factorization	711 selected sources; exact rational orbits through horizon four	Unequal successors in a fiber, or successor outside the verified image

The nonnegative coverage carrier includes bodies outside the signed dynamic carrier. The earlier first-reference successor scan is numerical; the two-source certificate establishes one exact inequality, while the full-domain census groups rational profiles by canonical integer keys. Reached-set replay starts from retained exact-orbit records rather than a fresh reconstruction of the microscopic producer.

T2. Nonnegative Coverage

Typed core excludes bodies absent from the curated annotation table. Relay strict admits them at endpoints; coverage inclusive also admits them as relays. A retained missing value within the table differs from an absent body.

Table T2. Supported and lifted depth-two triples by annotation policy.

Policy	Supported	Lifted	Incremental supported / lifted
Typed core	6,237	4,970	–
Relay strict	7,019	5,751	782 / 781
Coverage inclusive	7,831	6,125	812 / 374

Of the last increment's 812 candidates, 438 have no common microscopic relay. These counts do not belong to the signed dynamic test.

T3. Earlier Numerical Successor Scan

Within each source sector, the registered scan compared eligible unit sources with its first eligible node. Witness counts are neither all-pairs counts nor prevalence estimates.

Table T3. Paper XVI first-reference one-step observation defects.

Partition	Sectors	Witnesses	Largest L1 defect
superclass	28	141,296	0.005651574473947287
somaSide	4	163,435	0.0004874856094383272

T4. Coordinate-Resolved Census

For source sector E_i and target coordinate j, let n_{ijq} count sources with profile coordinate q. The exact number of unordered same-observation source pairs separated by j is

$$D_{ij} = \binom{N_i}{2} - \sum_q \binom{n_{ijq}}{2}.$$

Table T4. Absolute somaSide pair separations by target coordinate.

Target coordinate	Separated within-sector source pairs
L	5,008,763,423
R	4,993,360,210
missing-label	3,413,153,081
M	1,307,422,058

The denominator is 5,432,178,301 unordered within-sector pairs. Pairs separated by multiple coordinates appear in multiple rows, so these counts are not additive. Figure 2 gives the corresponding percentages.

T5. Common Reached-Set Support

Paper XVII selected 711 eligible sources across 25 nonempty source strata. The common comparison domain of the three history orders is $D_0 \times \{2, 3\}$.

Table T5. Exact history fibers on the common 1,422-window domain.

History order	Windows	Distinct values	Largest fiber	Failure fibers
0	1,422	1,228	51	0
1	1,422	1,228	51	0
2	1,422	1,228	51	0

The three partitions, not just their counts, agree. Five persistent cohorts have sizes 51, 25, 19, 4, and 3. Each of the $t = 2, 3, 4$ slices has 614 observation values. The $t = 2, 3$ slices are disjoint, while every $t = 4$ value escapes their combined verified image. Appendix A gives the distinct-state and drive-off argument.

Reproduction

The Paper XVIII artifact index binds the report's local release closure and its earlier source records.

The source packages are independently identified as Paper XVI and Paper XVII. The predictive-profile Dataset contains the exact full-domain census, source and focused controls, pinned derived carrier, and figure sources. Its verifier checks inventory and digests; optional replay recomputes the census.

The separate compact witness package contains the two-source certificate, checker, controls, and three Paper XVI metadata records. Its default check verifies compact arithmetic. Source-row re-extraction additionally needs the byte-pinned carrier from the Dataset. The Dataset DOI does not identify this ZIP. Neither replay independently reconstructs the original connectome pipeline.